

PHASE III – Solution Architecture, Proposed Solution & Problem-Solution Fit

Date	07 APRIL 2026
Team ID	69c378a771a975a7e423b048
Project Name	Streamlining Ticket Assignment for Efficient Support Operations
Maximum Marks	4 Marks

Solution Architecture

Architecture Overview

The solution is built on ServiceNow using a layered architecture:

1. **Presentation Layer**

- Service Portal / UI
- Ticket creation interface for users

1. **Application Layer**

- Assignment Rules
- Data Lookup Rules
- Flow Designer (automation logic)
- Business Rules (optional scripting)

1. **Data Layer**

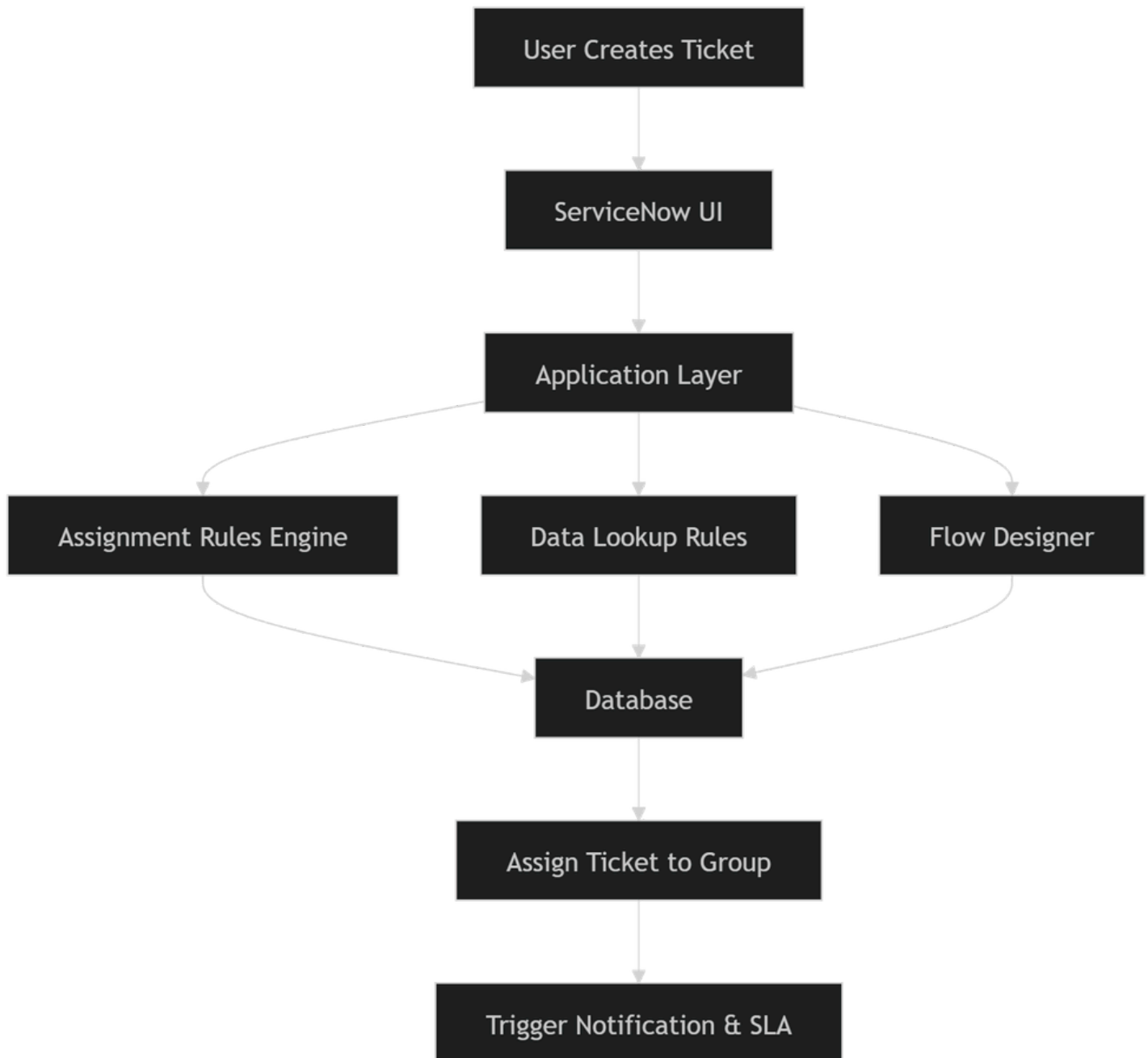
- Incident Table
- User Table
- Group Table
- SLA Table

Architecture Flow

Proposed Solution

Key Features

- Automated ticket assignment based on:
- Category
- Priority
- Location
- Reduction of manual intervention
- Real-time notifications to support teams
- SLA tracking and escalation
- Improved accuracy and efficiency



How It Works

1. User submits a ticket

- 2. System validates input data
- 3. Assignment rules determine the correct group
- 4. Data lookup rules auto-fill necessary fields
- 5. Ticket is assigned automatically
- 6. Notification is sent & SLA starts

Presentation Layer

- User interface for ticket submission (Service Portal / native UI)
- Dashboard for agents to track tickets and assignments

Application Layer

- **Assignment Rules:** Automatically route tickets based on category, priority, and location
- **Data Lookup Rules:** Auto-populate fields such as assignment group or impact
- **Flow Designer:** Orchestrates automated workflows, notifications, and SLA tracking
- **Business Rules / Scripts:** Optional advanced logic for edge cases

Data Layer

- **Incident Table:** Stores ticket details
- **User Table:** Contains information about users submitting tickets
- **Group Table:** Stores support team info
- **SLA Table:** Tracks service level metrics

 Problem–Solution Fit Template

Problem Identified	Proposed Solution	Expected Outcome
Manual ticket assignment	Automated Assignment Rules	Faster ticket routing
Incorrect ticket routing	Category-based logic	Improved accuracy
High workload on service desk	Automation & rule-based distribution	Balanced workload
SLA delays	Automatic SLA triggering	Better SLA compliance
Repeated ticket reassignment	Smart routing logic	Reduced rework

Phase 3 Outcome

- Designed scalable system architecture

- Defined automation-based solution
- Established clear problem–solution mapping
- Ready for implementation phase

In this phase, the focus is on designing a structured solution architecture and defining a proposed system to address the challenges identified earlier. The solution is built on ServiceNow, leveraging its ITSM capabilities to create an automated and efficient ticket assignment framework. The architecture consists of key components such as the user interface (where tickets are created), the application layer (where business logic like assignment rules and workflows are executed), and the database layer (where ticket data is stored and managed). These components work together to ensure seamless data flow and real-time processing of ticket assignments.

The proposed solution aims to eliminate manual intervention by introducing automation through Assignment Rules, Data Lookup Rules, and Flow Designer. When a ticket is created, the system automatically evaluates its attributes—such as category, priority, and location—and assigns it to the appropriate support group. Notifications are triggered instantly, and SLA tracking begins without delay. This ensures that tickets are handled efficiently, reducing response time and minimizing errors caused by manual routing.

To validate the effectiveness of the solution, a problem-solution fit approach is applied. The identified problem of delayed and inaccurate ticket assignment is directly addressed by automation and rule-based decision-making. The solution aligns closely with user needs by improving accuracy, reducing workload on service desk agents, and ensuring faster resolution times for end users. This alignment confirms that the proposed system not only solves the core problem but also enhances overall operational efficiency and service quality.

THANKING YOU....